

Practical Session 7

Task 1

Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

```
j_rakshitha@RAKSHITHA:~$ nano memory_addresses.c
j_rakshitha@RAKSHITHA:~$ gcc -Wall -Wextra -g memory_addresses.c -o memory_addresses
j_rakshitha@RAKSHITHA:~$ ./memory_addresses
Practical Session 7 - Task 1
=====
Code address      : 0x6293501521e9
Global variable   : 0x629350155010
Static global variable : 0x629350155014
Static local variable : 0x629350155018
Heap variable     : 0x629367f3f010
Stack variable    : 0x7ffe7dc4337c

Variable values:
Global           : 100
Static global    : 200
Static local     : 400
Heap             : 500
Stack            : 300

Heap memory freed successfully.
j_rakshitha@RAKSHITHA:~$ gcc -Wall -Wextra -g memory_addresses.c -o memory_addresses
j_rakshitha@RAKSHITHA:~$ ./memory_addresses
Practical Session 7 - Task 1
=====
Code address      : 0x63637f0b91e9
Global variable   : 0x63637f0bc010
Static global variable : 0x63637f0bc014
Static local variable : 0x63637f0bc018
Heap variable     : 0x636399237010
Stack variable    : 0x7ffda97fd44c

Variable values:
Global           : 100
Static global    : 200
Static local     : 400
Heap             : 500
Stack            : 300

Heap memory freed successfully.
j_rakshitha@RAKSHITHA:~$ |
```

```
j_rakshitha@RAKSHITHA: ~  
GNU nano 8.7.1 memory_addresses.c  
#include <stdio.h>  
#include <stdlib.h>  
  
/* Global variable */  
int global_var = 100;  
  
/* Static global variable */  
static int static_global_var = 200;  
  
/* Function represents the code segment */  
void sample_function(void)  
{  
    printf("Inside sample_function()\n");  
}  
  
int main(void)  
{  
    /* Local stack variable */  
    int stack_var = 300;  
  
    /* Static local variable */  
    static int static_local_var = 400;  
  
    /* Heap variable */  
    int *heap_var = malloc(sizeof(int));  
  
    if (heap_var == NULL)  
    {  
        perror("malloc");  
        return 1;  
    }  
  
    *heap_var = 500;  
  
    printf("Practical Session 7 - Task 1\n");  
    printf("=====\n");  
  
    printf("Code address      : %p\n", (void *)sample_function);  
    printf("Global variable    : %p\n", (void *)&global_var);  
    printf("Static global variable : %p\n", (void *)&static_global_var);  
    printf("Static local variable : %p\n", (void *)&static_local_var);  
    printf("Heap variable       : %p\n", (void *)heap_var);  
    printf("Stack variable      : %p\n", (void *)&stack_var);  
  
    printf("\nVariable values:\n");  
    printf("Global          : %d\n", global_var);  
    printf("Static global   : %d\n", static_global_var);  
    printf("Static local    : %d\n", static_local_var);  
    printf("Heap            : %d\n", *heap_var);  
    printf("Stack           : %d\n", stack_var);  
  
    free(heap_var);  
  
    printf("\nHeap memory freed successfully.\n");  
  
    return 0;  
}
```

Task 2

Make use of the `/proc/maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

```
j_rakshitha@RAKSHITHA:~$ ps
Static local      : 400
Heap              : 500
Stack            : 300

Heap memory freed successfully.
j_rakshitha@RAKSHITHA:~$ nano memory_addresses.c

j_rakshitha@RAKSHITHA:~$ nano proc_memory.c
j_rakshitha@RAKSHITHA:~$ gcc -Wall -Wextra -g proc_memory.c -o proc_memory
j_rakshitha@RAKSHITHA:~$ ./proc_memory
Practical Session 7 - Task 2
=====
Process PID: 1113

Global address : 0x61e04508d010
Heap address   : 0x61e072749010
Stack address  : 0x7fff5ea8372c

Process is running...
Press Enter to exit...

j_rakshitha@RAKSHITHA:~$ |

j_rakshitha@RAKSHITHA:~$ ps
PID TTY      TIME CMD
1121 pts/2    00:00:00 bash
1157 pts/2    00:00:00 ps
j_rakshitha@RAKSHITHA:~$ cat /proc/1113/maps
61e045089000-61e04508a000 r--p 00000000 00:30 40564      /home/j_rakshitha/
proc_memory
61e04508a000-61e04508b000 r-xp 00001000 00:30 40564      /home/j_rakshitha/
proc_memory
61e04508b000-61e04508c000 r--p 00002000 00:30 40564      /home/j_rakshitha/
proc_memory
61e04508c000-61e04508d000 r--p 00002000 00:30 40564      /home/j_rakshitha/
proc_memory
61e04508d000-61e04508e000 rw-p 00003000 00:30 40564      /home/j_rakshitha/
proc_memory
61e072749000-61e07274a000 rw-p 00000000 00:00 0          [heap]
7a3368200000-7a3368228000 r--p 00000000 00:30 13647      /usr/lib/x86_64-li
nux-gnu/libc.so.6
7a3368228000-7a33683c0000 r-xp 00028000 00:30 13647      /usr/lib/x86_64-li
nux-gnu/libc.so.6
7a33683c0000-7a336840e000 r--p 001c0000 00:30 13647      /usr/lib/x86_64-li
nux-gnu/libc.so.6
7a336840e000-7a3368412000 r--p 0020d000 00:30 13647      /usr/lib/x86_64-li
nux-gnu/libc.so.6
7a3368412000-7a3368414000 rw-p 00211000 00:30 13647      /usr/lib/x86_64-li
nux-gnu/libc.so.6
7a3368414000-7a3368421000 rw-p 00000000 00:00 0
7a33685b4000-7a33685b7000 rw-p 00000000 00:00 0
7a33685b7000-7a33685bf000 rw-p 00000000 00:00 0
7a33685bf000-7a33685c3000 r--p 00000000 00:00 0          [vvar]
7a33685c3000-7a33685c5000 r--p 00000000 00:00 0          [vvar_vclock]
7a33685c5000-7a33685c7000 r-xp 00000000 00:00 0          [vdso]
7a33685c7000-7a33685c8000 r--p 00000000 00:30 13644      /usr/lib/x86_64-li
nux-gnu/ld-linux-x86-64.so.2
7a33685c8000-7a33685f7000 r-xp 00001000 00:30 13644      /usr/lib/x86_64-li
nux-gnu/ld-linux-x86-64.so.2
7a33685f7000-7a3368602000 r--p 00030000 00:30 13644      /usr/lib/x86_64-li
nux-gnu/ld-linux-x86-64.so.2
7a3368602000-7a3368604000 r--p 0003b000 00:30 13644      /usr/lib/x86_64-li
nux-gnu/ld-linux-x86-64.so.2
7a3368604000-7a3368605000 rw-p 0003d000 00:30 13644      /usr/lib/x86_64-li
nux-gnu/ld-linux-x86-64.so.2
7a3368605000-7a3368606000 rw-p 00000000 00:00 0
7fff5ea60000-7fff5ea85000 rw-p 00000000 00:00 0          [stack]
j_rakshitha@RAKSHITHA:~$ pmmap -x 1113
1113: ./proc_memory
Address      Kbytes    RSS    Dirty Mode Mapping
000061e045089000    4         4      0 r----- proc_memory
000061e04508a000    4         4      0 r-x----- proc_memory
```

```
j_rakshitha@RAKSHITHA: ~
1113: ./proc_memory
Address      Kbytes      RSS      Dirty Mode Mapping
000061e045089000      4          4          0 r---- proc_memory
000061e04508a000      4          4          0 r-x-- proc_memory
000061e04508b000      4          4          0 r---- proc_memory
000061e04508c000      4          4          4 r---- proc_memory
000061e04508d000      4          4          4 rw--- proc_memory
000061e072749000     132          4          4 rw--- [ anon ]
00007a3368200000     160         160          0 r---- libc.so.6
00007a3368228000    1632       1120          0 r-x-- libc.so.6
00007a33683c0000     312         184          0 r---- libc.so.6
00007a336840e000      16          16         16 r---- libc.so.6
00007a3368412000      8           8          8 rw--- libc.so.6
00007a3368414000     52          20         20 rw--- [ anon ]
00007a33685b4000     12           8          8 rw--- [ anon ]
00007a33685bd000      8           4          4 rw--- [ anon ]
00007a33685bf000     16           0          0 r---- [ anon ]
00007a33685c3000      8           0          0 r---- [ anon ]
00007a33685c5000      8           8          0 r-x-- [ anon ]
00007a33685c7000      4           4          0 r---- ld-linux-x86-64.so.2
00007a33685c8000     188         188          0 r-x-- ld-linux-x86-64.so.2
00007a33685cf7000     44          36          0 r---- ld-linux-x86-64.so.2
00007a3368602000      8           8          8 r---- ld-linux-x86-64.so.2
00007a3368604000      4           4          4 rw--- ld-linux-x86-64.so.2
00007a3368605000      4           4          4 rw--- [ anon ]
00007fff5ea64000     132          12         12 rw--- [ stack ]
-----
total kB      2768      1808      96
j_rakshitha@RAKSHITHA:~$ cat /proc/1113/maps
61e045089000-61e04508a000 r--p 00000000 08:30 40564 /home/j_rakshitha/
proc_memory
61e04508a000-61e04508b000 r-xp 00001000 08:30 40564 /home/j_rakshitha/
proc_memory
61e04508b000-61e04508c000 r--p 00002000 08:30 40564 /home/j_rakshitha/
proc_memory
61e04508c000-61e04508d000 r--p 00002000 08:30 40564 /home/j_rakshitha/
proc_memory
61e04508d000-61e04508e000 rw-p 00003000 08:30 40564 /home/j_rakshitha/
proc_memory
61e072749000-61e07276a000 rw-p 00000000 00:00 0 [heap]
7a3368200000-7a3368228000 r--p 00000000 08:30 13647 /usr/lib/x86_64-li
nux-gnu/libc.so.6
7a3368228000-7a33683c0000 r-xp 00028000 08:30 13647 /usr/lib/x86_64-li
nux-gnu/libc.so.6
7a33683c0000-7a336840e000 r--p 001c0000 08:30 13647 /usr/lib/x86_64-li
nux-gnu/libc.so.6
7a336840e000-7a3368412000 r--p 0020d000 08:30 13647 /usr/lib/x86_64-li
nux-gnu/libc.so.6
7a3368412000-7a3368414000 rw-p 00211000 08:30 13647 /usr/lib/x86_64-li
```

```

j_rakshitha@RAKSHITHA: ~
proc_memory
61e04508d000-61e04508e000 rw-p 00003000 08:30 40564 /home/j_rakshitha/
proc_memory
61e072749000-61e07276a000 rw-p 00000000 00:00 0 [heap]
7a3368200000-7a3368228000 r--p 00000000 08:30 13647 /usr/lib/x86_64-li
nux-gnu/libc.so.6
7a3368228000-7a33683c0000 r-xp 00028000 08:30 13647 /usr/lib/x86_64-li
nux-gnu/libc.so.6
7a33683c0000-7a336840e000 r--p 001c0000 08:30 13647 /usr/lib/x86_64-li
nux-gnu/libc.so.6
7a336840e000-7a3368412000 r--p 0020d000 08:30 13647 /usr/lib/x86_64-li
nux-gnu/libc.so.6
7a3368412000-7a3368414000 rw-p 00211000 08:30 13647 /usr/lib/x86_64-li
nux-gnu/libc.so.6
7a3368414000-7a3368421000 rw-p 00000000 00:00 0
7a33685b4000-7a33685b7000 rw-p 00000000 00:00 0
7a33685bd000-7a33685bf000 rw-p 00000000 00:00 0
7a33685bf000-7a33685c3000 r--p 00000000 00:00 0 [vvar]
7a33685c3000-7a33685c5000 r--p 00000000 00:00 0 [vvar_vclock]
7a33685c5000-7a33685c7000 r-xp 00000000 00:00 0 [vdso]
7a33685c7000-7a33685c8000 r--p 00000000 08:30 13644 /usr/lib/x86_64-li
nux-gnu/ld-linux-x86-64.so.2
7a33685c8000-7a33685f7000 r-xp 00001000 08:30 13644 /usr/lib/x86_64-li
nux-gnu/ld-linux-x86-64.so.2
7a33685f7000-7a3368602000 r--p 00030000 08:30 13644 /usr/lib/x86_64-li
nux-gnu/ld-linux-x86-64.so.2
7a3368602000-7a3368604000 r--p 0003b000 08:30 13644 /usr/lib/x86_64-li
nux-gnu/ld-linux-x86-64.so.2
7a3368604000-7a3368605000 rw-p 0003d000 08:30 13644 /usr/lib/x86_64-li
nux-gnu/ld-linux-x86-64.so.2
7a3368605000-7a3368606000 rw-p 00000000 00:00 0
7fff5ea64000-7fff5ea85000 rw-p 00000000 00:00 0 [stack]
j_rakshitha@RAKSHITHA:~$

```

j_rakshitha@RAKSHITHA: ~

```
GNU nano 8.7.1 proc_memory.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 100;

int main(void)
{
    int stack_var = 200;

    int *heap_var = malloc(sizeof(int));

    if (heap_var == NULL)
    {
        perror("malloc");
        return 1;
    }

    *heap_var = 300;

    printf("Practical Session 7 - Task 2\n");
    printf("=====\n");
    printf("Process PID: %d\n\n", getpid());

    printf("Global address : %p\n", (void *)&global_var);
    printf("Heap address   : %p\n", (void *)heap_var);
    printf("Stack address  : %p\n", (void *)&stack_var);

    printf("\nProcess is running...\n");
    printf("Press Enter to exit...\n");

    fflush(stdout);

    getchar();

    free(heap_var);

    return 0;
}
```